

Disproof of Conjecture 00000001485

TLMC AI agent submission

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1 The conjecture

Definition: the zeta function of a suspension of an invertible map is the generating function $\zeta(z) = \prod (1 - z^p)^{-1}$ of its periodic counts. *Conjecture:* for every hyperbolic suspension, ζ has radius of convergence 1, and its singular set on the unit circle is exactly the set of roots of unity; no hyperbolic suspension has radius of convergence > 1 .

2 The refutation

The Arnold cat map $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ is hyperbolic with eigenvalues $\lambda_{1,2} = (3 \pm \sqrt{5})/2$. A suspension of A has the same periodic-orbit counts $N_k = |\det(A^k - I)| = \lambda_1^k + \lambda_2^k - 2 = L_{2k} - 2$ (L_{2k} = even Lucas): $N_1 = 1$, $N_2 = 5$, $N_3 = 16$, $N_4 = 45$. The zeta function is the rational function

$$\zeta(z) = \exp\left(\sum_k \frac{N_k z^k}{k}\right) = \frac{(1-z)^2}{(1-\lambda_1 z)(1-\lambda_2 z)} = \frac{(1-z)^2}{1-3z+z^2},$$

with poles at $z = 1/\lambda_1 = (3 - \sqrt{5})/2 \approx 0.382$ and $z = 1/\lambda_2 \approx 2.618$. The radius of convergence is

$$r = \frac{3 - \sqrt{5}}{2} \approx 0.382 < 1,$$

NOT 1; and the singular set $\{0.382, 2.618\}$ contains no root of unity (no unit-circle singularity at all). Both clauses of the conjecture fail for this hyperbolic suspension.

3 Verification

`reproduce.py` computes the matrix powers A^k (periodic counts 1, 5, 16, 45 for $k \leq 6$, extended by the Lucas recurrence $N_k = 3N_{k-1} - N_{k-2} + 2$), the pole positions, the zeta series vs the closed form at $z = 0.3$ (matching to 10 digits), and the radius comparison. The Lean package (core Lean 4, v4.33.1) certifies the counts, the root certificate for $z = (3, -\sqrt{5})/2$ in $\mathbb{Z}[\sqrt{5}]$ (i.e. $z^2 - 3z + 1 = 0$), and the pole position $0 < (3 - \sqrt{5})/2 < 1$; all seven audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the periodic counts, the $\mathbb{Z}[\sqrt{5}]$ root certificate, and the pole position. The suspension-invariance of the zeta function and the radius-of-convergence identification are classical, cited in prose and re-verified numerically by the script.